QC.4) Tune PI controller:

We'll use the IMC-PID tuning rules →

Our model:

$K_p = 0.664$
$\tau_p = 759\,s$
$\theta_p = 6\,s$

$\tau_I = \tau_p + \frac{\theta_p}{2} = 762\,s$
$\tau_c = \max(0.1\tau_p, 0.8\theta_p) = 75.9\,s$
$K_c = \frac{\tau_c}{K_p(\tau_c)} = 12.99$

Manufacturer model:

$K_p = 0.5$
$\tau_p = 120\,s$
$\theta_p = 10\,s$

$\tau_I = 125\,s$
$\tau_c = 12\,s$
$K_c = 14.71$

$\tau_D = 0$ for PI